

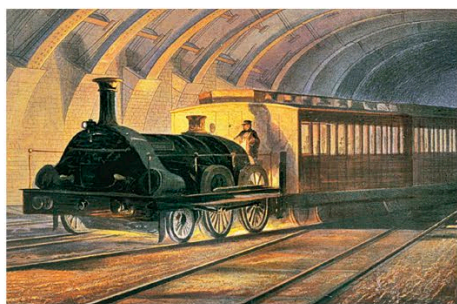
On the tracks

Since the world's first railroad opened between Stockton and Darlington in the UK in 1825, train technology has come a long way. For well over a century, steam locomotives dominated the railroads, but in the last 50 years, diesel and electric locomotives have almost entirely replaced them.

First subway

- **What?** Metropolitan Railway
- **Who?** John Fowler
- **Where and when?** UK, 1863

The problem of building railroads in crowded cities was solved by putting them underground. The world's first was the Metropolitan Railway, which opened in January 1863 in London. It had gas-lit wooden cars hauled by steam locomotives, but smoke was a choking hazard in the confined tunnels.



Fastest steam locomotive

- **What?** *Mallard*
- **Who?** Sir Nigel Gresley
- **Where and when?** UK, 1938



Deep-tube electric railroad

- **What?** City and South London Railway
- **Who?** James Henry Greathead
- **Where and when?** UK, 1890

Electric underground railroads are now the way people get around in many of the world's biggest cities. The first one was built in London in 1890, between the city and Stockwell (a suburb of London). The tunnels were so small and the cars so tiny that the cars were nicknamed "padded cells."



Diesel-electric locomotive

- **What?** EMD FT
- **Who?** General Motors, EMD
- **Where and when?** US, 1939

By the time this powerful locomotive began its run, diesel locomotives were replacing the steam engines. Unlike electric locomotives, which are usually powered by overhead electric lines, diesel-electric engines could be operated on existing tracks with no modification. The "F" stood for Fourteen Hundred horsepower, and the "T" for Twin, because FTs were always sold in pairs.

WOW!

In 2016, an experimental Japanese maglev train hit a top speed of 374 mph (603 km/h).

